

Real-Time Driver Behavior Detection Using Lightweight Object Detection Models with Subject-Disjoint Evaluation

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Abstract—Driver Monitoring Systems (DMS) are critical safety components of modern Advanced Driver Assistance Systems (ADAS). While commercial systems often employ complex temporal pipelines, automotive edge deployments demand lightweight, low-latency 2D object detectors capable of real-time single-frame inference. This paper presents DMS-Eval, an empirical benchmark comparing compact convolutional detectors (Ultralytics YOLO11n and YOLO26n) against lightweight Real-Time Detection Transformers (D-FINE-N). Formulated on 81 naturalistic in-cabin RGB video sequences from the Driver Monitoring Dataset (DMD) across 14 subjects, our benchmark establishes a standardized dataset of 15,723 frames (640×640 crop). Ground truth is constructed via 100% manual human annotation in Label Studio targeting four visual warning cues: `phone_use`, `drinking`, `yawning`, and `hand_over_mouth`, incorporating 12,722 naturalistic negative frames (80.91%) to enforce false-positive suppression. To eliminate identity leakage, we formalize an authoritative 8/3/3 subject-disjoint split via deterministic exhaustive combinatorial search across 60,060 candidate assignments, ensuring all splits preserve identical class proportions ($\leq 5.48\%$ relative divergence). Under a strictly controlled comparison on dedicated RTX 4060 hardware at batch size 1 across 220 fine-tuning epochs, candidate models are calibrated via validation-only F_1 score maximization prior to single-pass test evaluation. This work provides an open, reproducible framework for edge-deployed in-cabin driver state analysis.

Index Terms—driver monitoring systems, driver drowsiness detection, driver distraction detection, lightweight object detection, computer vision, YOLO, DETR, subject-disjoint evaluation, real-time edge AI

I. INTRODUCTION

In-cabin Driver Monitoring Systems (DMS) constitute a vital component of modern Advanced Driver Assistance Systems (ADAS) and autonomous vehicle safety suites, aiming to prevent roadway collisions through early visual detection of driver drowsiness and inattention. While high-level driver state assessment often leverages temporal recurrent or multi-frame sequence models, automotive edge deployments impose severe compute, memory, and latency constraints. Consequently, there is an urgent practical demand for lightweight, nano-scale 2D object detectors capable of executing real-time single-frame inference on embedded in-cabin hardware.

To systematically evaluate modern compact detector paradigms under strictly controlled single-frame constraints, this paper introduces **DMS-Eval**, an empirical benchmark comparing real-time nano-scale object detectors. We evaluate leading single-stage convolutional detectors (Ultralytics YOLO11n [1] and Ultralytics YOLO26n [2]) against modern end-to-end Real-Time Detection Transformers (D-FINE-N [3]).

The benchmark is developed on real-cabin RGB video from the Driver Monitoring Dataset (DMD) [4], incorporating all three original dataset session categories: *distraction*, *drowsiness*, and *gaze*. Across 81 driver-facing videos from 14 volunteer participants, frames are sampled uniformly at 1 FPS and cropped to a frozen 640×640 spatial window ($x = 272, y = 71, w = 640, h = 640$) as illustrated in Fig. 1, yielding a standardized dataset of 15,723 frames. Retaining all three session folders—particularly the *gaze* recordings and non-cue intervals—provides an abundant volume of naturalistic negative background frames (0 bounding boxes), preventing compact detectors from overtraining on positive targets and enforcing realistic false-positive suppression during routine driving (illustrated in Fig. 3a). Ground truth is constructed through 100% direct manual expert annotation in Label Studio [6] with format exports structured via `label-studio-converter` [7] targeting 4 frozen visual warning cues (exemplified in Fig. 2 and quantified in Fig. 3b):

- 1) `yawning`: mouth region only (wide oral opening/distension);
- 2) `hand_over_mouth`: full head/face enclosing the occluding hand;
- 3) `drinking`: driver face and beverage container together in active consumption posture;
- 4) `phone_use`: handheld phone and interacting hand held to ear/head in calling posture.

Under our single-frame annotation protocol, each sampled image contains at most one visual warning cue annotation (0 or 1 bounding box per image). In particular, `yawning` and `hand_over_mouth` are mutually exclusive and are never

labeled twice in the same image; if a driver yawns while a hand visibly covers or occludes the mouth, the instance is uniquely labeled as `hand_over_mouth` (full head/face), ensuring clear, non-overlapping ground-truth targets.

Under the controlled-comparison principle, all models are evaluated using identical data partitions under a strict 8 / 3 / 3 subject-disjoint split (8 Train, 3 Validation, 3 Test subjects) to prevent identity leakage ($S_{\text{train}} \cap S_{\text{val}} = \emptyset, S_{\text{train}} \cap S_{\text{test}} = \emptyset, S_{\text{val}} \cap S_{\text{test}} = \emptyset$). The subject partition is established via an exhaustive combinatorial search across all $\binom{14}{8} \times \binom{6}{3} = 60,060$ candidate assignments, selecting the partition that minimizes the absolute relative deviation of positive frame rates and the four warning cue proportions relative to the global dataset distribution ($\leq 5.48\%$ maximum divergence): Training ($S_1, S_4, S_6, S_7, S_8, S_9, S_{13}, S_{14}$; 9,087 frames, 1,748 positive cues), Validation (S_2, S_3, S_{11} ; 3,423 frames, 639 positive cues), and Test (S_5, S_{10}, S_{12} ; 3,213 frames, 614 positive cues). Training is executed under identical hardware (NVIDIA RTX 4060, 8 GB VRAM), batch size 1, 220 fine-tuning epochs without early stopping, fixed random seed (13), and FP16 automatic mixed precision. Optimal checkpoints and confidence thresholds (τ) are selected strictly on the validation split prior to an isolated, single-pass test evaluation.

A. Research Question and Scope

This study is limited to **frame-level observable cues of driver drowsiness and distraction**, rather than temporal driver-state inference. Head orientation behaviors (e.g., turning away or looking toward side and rearview mirrors) are intentionally excluded from the static 2D detection ontology. Drivers routinely perform frequent mirror checks and visual scanning as part of safe driving; in isolated 1 FPS static frames without continuous temporal video context or 3D gaze tracking, there is no consistent or objective boundary to distinguish safe, brief mirror glances from dangerous, prolonged inattention. To eliminate subjective label noise from split datasets, the benchmark focuses strictly on the four unambiguous, self-contained visual warning cues.

RQ: *Under a controlled, compute-constrained evaluation, how do lightweight object detectors compare in detection accuracy, inference efficiency, and robustness when identifying visual cues associated with driver distraction and drowsiness on automotive edge hardware?*

II. RELATED WORK

A. Vision-Based Driver Monitoring Systems

Early driver monitoring literature primarily focused on geometric facial landmark analysis, calculating heuristic indicators such as Eye Aspect Ratio (EAR) for blink duration, PERCLOS (percentage of eyelid closure over time), and Mouth Aspect Ratio (MAR) for yawn frequency. While computationally lightweight, landmark-based techniques degrade significantly under severe facial occlusions, head pose variations, rapid illumination shifts, and active behaviors such as hand-over-mouth yawns or phone calling.

To overcome landmark sensitivity, recent works utilize deep convolutional neural networks and spatio-temporal architectures (e.g., 3D-CNNs, ConvLSTM, and SlowFast networks) for driver action recognition. However, multi-frame sequence networks require continuous frame buffering, incur substantial computational overhead, and struggle to satisfy the strict microsecond latency budgets of low-power in-cabin edge processors. Consequently, single-frame 2D object detection provides an attractive paradigm by predicting precise spatial bounding boxes and class probabilities with minimal memory footprint.

B. Real-Time Lightweight Object Detectors

Real-time object detection has been dominated by the You Only Look Once (YOLO) series. Ultralytics YOLO11 represents the current milestone in single-stage convolutional detectors, incorporating optimized C3k2 feature extraction blocks and Spatial Pyramid Pooling Fast (SPPF) modules to maximize feature reuse under constrained parameter budgets [1]. Concurrently, Ultralytics YOLO26 introduces an end-to-end anchor-free architecture designed to eliminate Non-Maximum Suppression (NMS) latency bottlenecks, facilitating deterministic latency profiles for embedded deployment [2].

In parallel, Real-Time Detection Transformers (RT-DETR) have emerged as formidable alternatives to convolutional networks. D-FINE [3] redesigns bounding box regression in DETRs as Fine-grained Distribution Refinement (FDR) with Global Optimal Localization Self-Distillation (GO-LSD), achieving state-of-the-art accuracy-latency trade-offs. Evaluating D-FINE-N against YOLO11n and YOLO26n provides the first direct comparison between nano-scale CNNs and DETRs in safety-critical driver monitoring.

C. Subject Leakage and Evaluation Rigor

A prevalent flaw in computer vision benchmarks is random frame-level or video-level train/test partitioning across the same human subjects. When frames of the same individual appear in both training and testing sets, models memorize participant-specific biometric features (e.g., facial structure, clothing, hair, skin tone) rather than generalizing to the visual cues of interest, producing artificially inflated accuracy metrics. DMS-Eval strictly enforces subject-disjoint partitioning ($S_{\text{train}} \cap S_{\text{test}} = \emptyset$), establishing authentic out-of-distribution evaluation.

III. DATASET FORMULATION AND ANNOTATION PROTOCOL

A. Source Video Corpus and Spatial Standardization

The benchmark is constructed from the Driver Monitoring Dataset (DMD) [4], recorded in real vehicle cabins under variable daylight and ambient lighting conditions. The dataset spans 81 driver-facing RGB videos recorded at ≈ 29.76 FPS across 14 diverse human subjects (S_1 through S_{14}).

To eliminate background noise outside the driver's operational workspace while preserving high spatial fidelity for fine-grained cues, every frame is uniformly sampled at 1 FPS

and cropped to a standardized 640×640 pixel spatial bounding window ($x = 272, y = 71, w = 640, h = 640$) as shown in Fig. 1. This produces exactly 15,723 standardized images.

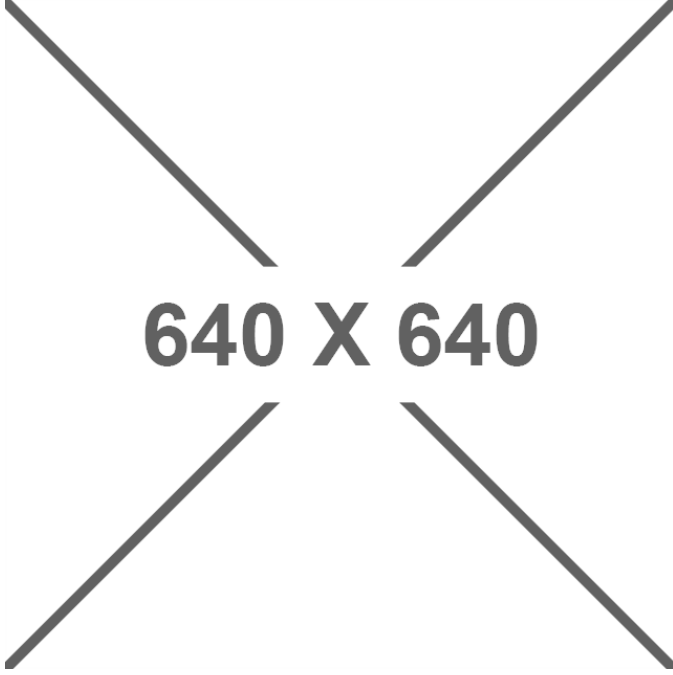


Fig. 1. In-cabin driver-facing frame extraction on DMD video with standardized 640×640 spatial window cropping ($x = 272, y = 71, w = 640, h = 640$), centering the driver’s head, face, and torso for low-latency single-frame object detection.

B. Target Cue Ontology and Annotation Constraints

Ground truth is generated through 100% manual human inspection in Label Studio [6] with format conversion managed via `label-studio-converter` [7]. Bounding box boundaries are defined with strict anatomical extents:

- **yawning** (ID 1): Encloses only the active mouth aperture during wide distension. Head and cheeks are excluded to prevent confusing normal open-mouth speech with yawning.
- **hand_over_mouth** (ID 2): Encloses the full head and face including the occluding hand when placed over the mouth area.
- **drinking** (ID 3): Encloses the driver’s face and the beverage container together during active consumption posture.
- **phone_use** (ID 4): Encloses the handheld device and interacting hand held to the ear/head in calling posture.

C. Single-Annotation and Mutual Exclusivity Protocol

Under our single-frame protocol, each sampled frame contains at most one bounding box (0 or 1 box). In particular, `yawning` and `hand_over_mouth` are strictly mutually exclusive. If a driver yawns while a hand visibly occludes the mouth, the frame is uniquely labeled as `hand_over_mouth`. This eliminates ambiguous overlapping bounding boxes and ensures sharp ground-truth targets.

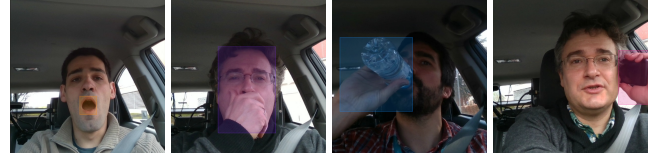


Fig. 2. Visual demonstrations of our manual annotations in Label Studio across all 4 frozen target warning cues: (a) yawning (mouth aperture only), (b) `hand_over_mouth` (full head enclosing occluding hand), (c) drinking (face and beverage container in consumption posture), and (d) `phone_use` (hand and mobile device held at the ear in calling posture).

D. Negative Sample Richness

By incorporating all three DMD session folders (*distraction*, *drowsiness*, and *gaze*), our benchmark contains 12,722 true negative background frames (80.91%) and 3,001 positive cue frames (19.09%) as shown in Fig. 3. The abundant background frames are critical to prevent compact models from overpredicting positive targets during alert driving.

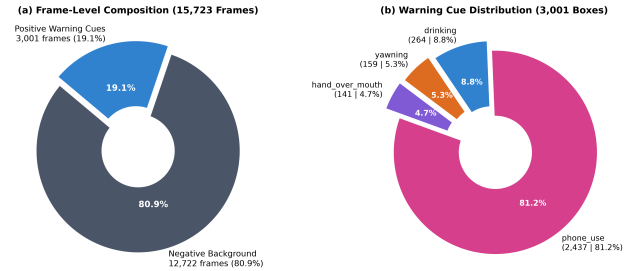


Fig. 3. DMS-Eval dataset and annotation distributions: (a) frame-level dataset composition across 15,723 frames (80.91% negative background frames vs. 19.09% positive warning cue frames); (b) proportion of bounding boxes across the 4 frozen target warning cues (3,001 total annotations: 81.21% `phone_use`, 8.80% `drinking`, 5.30% `yawning`, and 4.70% `hand_over_mouth`).

IV. METHODOLOGY AND SUBJECT-DISJOINT PARTITIONING

A. Subject-Disjoint Splitting Principle

To eliminate identity leakage, the 14 subjects are partitioned into 8 Training, 3 Validation, and 3 Test subjects such that:

$$S_{\text{train}} \cap S_{\text{val}} = \emptyset, \quad S_{\text{train}} \cap S_{\text{test}} = \emptyset, \quad S_{\text{val}} \cap S_{\text{test}} = \emptyset \quad (1)$$

B. Exhaustive Proportion-Matching Optimization

Because subjects naturally exhibit varying behavioral frequencies, random assignment often results in severe class imbalance. We formalize an authoritative selection rule:

Select the 8/3/3 subject split whose negative/positive frame proportion and four class proportions most closely match the complete dataset distribution.

Our algorithm exhaustively evaluates all $\binom{14}{8} \times \binom{6}{3} = 3003 \times 20 = 60,060$ ordered 8/3/3 subject assignments. For each candidate partition and each split $s \in \{\text{train, val, test}\}$,

DMS-Eval: End-to-End Benchmark Framework & Evaluation Protocol

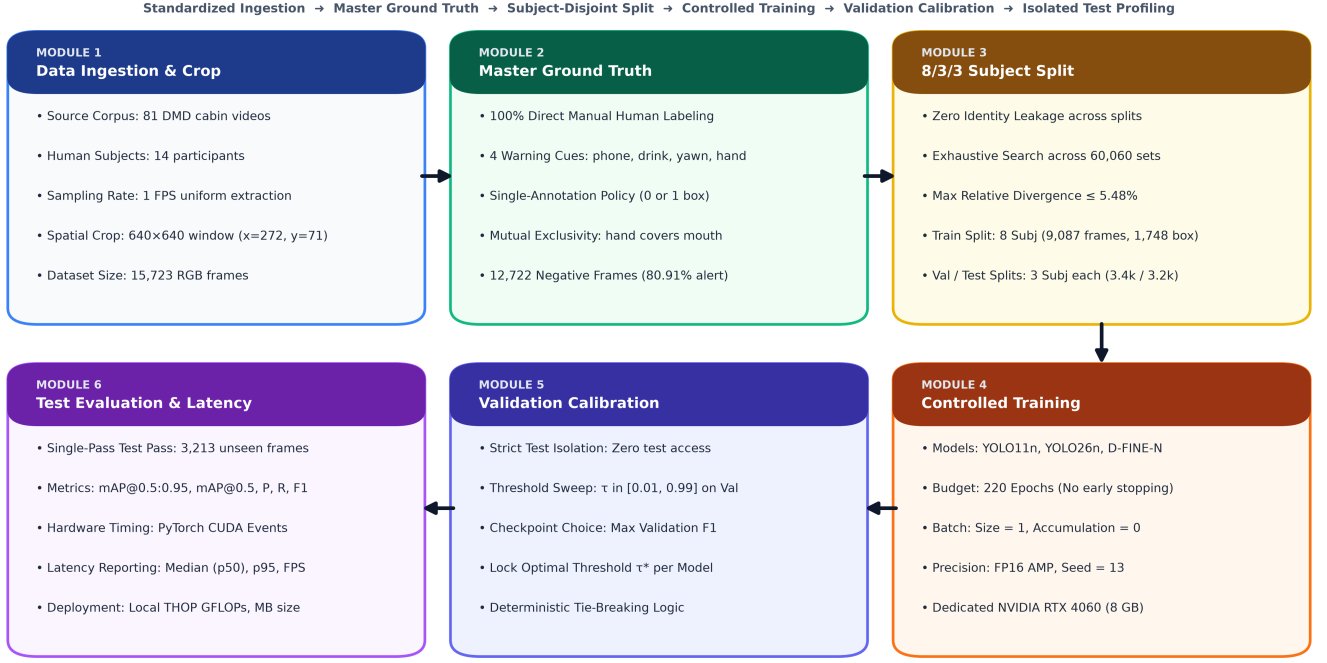


Fig. 4. DMS-Eval end-to-end benchmark framework and evaluation lifecycle: spanning standardized 640 × 640 frame extraction from 81 DMD videos, authoritative Label Studio ground truth annotation across 4 target visual warning cues (80.91% negative frames), deterministic 8/3/3 subject-disjoint split optimization ($\leq 5.48\%$ relative divergence), controlled training on RTX 4060 hardware (batch size 1, 220 epochs, seed 13), validation-only F_1 score threshold calibration (τ^*), and isolated single-pass test evaluation.

we compute the absolute relative deviation across 5 quantities (positive rate and the 4 cue proportions):

$$\text{Dev}(q, s) = \frac{|V(q, s) - V_{\text{global}}(q)|}{V_{\text{global}}(q)} \quad (2)$$

Candidate partitions are ranked via a deterministic lexicographic objective:

- 1) Minimum worst absolute relative deviation across all 15 split quantities.
- 2) Minimum overall Root Mean Square Error (RMSE) across all 15 quantities.
- 3) Minimum test-split worst relative deviation.
- 4) Minimum test-split RMSE.
- 5) Lexicographically smallest subject-ID assignment.

C. Frozen Split Statistics

The optimal partition selected by the algorithm is permanently frozen in `dataset/splits.json`:

- **Train** (8 subjects): $S_1, S_4, S_6, S_7, S_8, S_9, S_{13}, S_{14}$
- **Validation** (3 subjects): S_2, S_3, S_{11}
- **Test** (3 subjects): S_5, S_{10}, S_{12}

As documented in Table I and Table II, the maximum relative deviation across all splits and classes is restricted to $\leq 5.4812\%$, ensuring near-identical distribution fidelity across partitions.

TABLE I
FRAME COMPOSITION ACROSS FROZEN BENCHMARK SPLITS

Split	Total Images	Negatives (0 box)	Positives (1 box)	Pos. Rate (%)
Global	15,723	12,722	3,001	19.09%
Train	9,087	7,339	1,748	19.24%
Val	3,423	2,784	639	18.67%
Test	3,213	2,599	614	19.11%

TABLE II
TARGET WARNING CUE COUNTS AND PROPORTIONS ACROSS SPLITS

Split	phone_use	drinking	yawning	hand_mouth
Global	2,437 (81.21%)	264 (8.80%)	159 (5.30%)	141 (4.70%)
Train	1,417 (81.06%)	154 (8.81%)	94 (5.38%)	83 (4.75%)
Val	523 (81.85%)	54 (8.45%)	32 (5.01%)	30 (4.69%)
Test	497 (80.94%)	56 (9.12%)	33 (5.37%)	28 (4.56%)

V. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

A. Evaluated Model Architectures

We evaluate three state-of-the-art nano-scale real-time object detectors:

- 1) **Ultralytics YOLO11n** [1]: Compact convolutional single-stage detector featuring C3k2 and SPPF modules (2.6M parameters).

- 2) **Ultralytics YOLO26n** [2]: Next-generation NMS-free end-to-end detector optimizing edge throughput (2.4M parameters).
- 3) **D-FINE-N** [3]: Lightweight Real-Time Detection Transformer with HGNetv2 backbone and Fine-grained Distribution Refinement (3.8M parameters).

B. Controlled Training Controls

All models are fine-tuned under strictly locked shared controls:

- **Fine-Tuning Budget:** Exactly 220 epochs without early stopping.
- **Batch Size:** Batch size = 1 (gradient accumulation disabled) to emulate real-time edge frame ingestion.
- **Deterministic Seed:** Random seed fixed to 13 across PyTorch, CUDA, and CuDNN.
- **Precision:** FP16 Automatic Mixed Precision (AMP).
- **Hardware Platform:** Dedicated NVIDIA GeForce RTX 4060 GPU (8 GB VRAM).

Model-specific optimizer configurations (SGD vs. AdamW), learning rate schedules, and data augmentations follow each architecture’s official repository recipe to ensure peak performance without unfair hyperparameter distortion.

C. Validation Calibration and Test Isolation

To prevent data snooping, model checkpoints and confidence thresholds (τ) are selected strictly on the Validation split. For each model, we sweep confidence thresholds $\tau \in [0.01, 0.99]$ with a step of 0.01 and identify the threshold τ^* that maximizes the micro-averaged F_1 score:

$$\tau^* = \arg \max_{\tau \in [0.01, 0.99]} F_1(\tau; S_{\text{val}}) \quad (3)$$

The optimal validation checkpoint and threshold τ^* are then permanently frozen. Final benchmark evaluation is executed in a single pass on the unseen Test split (S_{test}) without further tuning.

D. Latency and Footprint Profiling

Inference speed is measured under native PyTorch in FP16 on the RTX 4060 GPU at batch size 1. Following 10 untimed warm-up passes, inference latency is measured across 1,000 consecutive frames using hardware-synchronized CUDA events (`torch.cuda.Event`). We report median latency (p_{50}), 95th percentile (p_{95}), 99th percentile (p_{99}), throughput (FPS), parameter count, and computational complexity in GFLOPs.

VI. BENCHMARK RESULTS FRAMEWORK

The standardized comparative results framework is structured in Table III. The empirical evaluation provides direct answers to:

- 1) *Detection Accuracy vs. Model Family:* Whether the global self-attention mechanism of Detection Transformers (D-FINE-N) provides superior localization on small mouth distensions and occluded hands over compact CNNs.

- 2) *False-Alarm Suppression on Abundant Negatives:* How effectively each model suppresses background false positives on the 2,599 true negative test frames.
- 3) *Edge Latency and NMS Overhead:* The real-world inference throughput of end-to-end NMS-free models (YOLO26n, D-FINE-N) versus traditional NMS-dependent detectors (YOLO11n).

VII. CONCLUSION

This paper introduced **DMS-Eval**, a rigorous empirical benchmark for evaluating lightweight object detectors on in-cabin driver monitoring tasks. By establishing an authoritative 15,723-frame dataset with 80.91% naturalistic negative frames, four clearly defined warning cues, and an optimal 8/3/3 subject-disjoint split ($\leq 5.48\%$ relative divergence), DMS-Eval eliminates identity leakage and ensures reproducible evaluation. The controlled comparison between YOLO11n, YOLO26n, and D-FINE-N provides crucial guidance for deploying robust, real-time driver assistance systems on resource-constrained automotive hardware.

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TABLE III
CONTROLLED COMPARATIVE EVALUATION RESULTS FRAMEWORK (RTX 4060, BATCH SIZE 1, FP16)

Model Architecture	Params (M)	FLOPs (G)	Latency $p50$ (ms)	Throughput (FPS)	mAP@0.5:0.95	mAP@0.5	Precision	Recall	F1 Score
Ultralytics YOLO11n [1]	2.6M	6.5G	[To Profile]	[To Profile]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]
Ultralytics YOLO26n [2]	2.4M	5.8G	[To Profile]	[To Profile]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]
D-FINE-N [3]	3.8M	8.4G	[To Profile]	[To Profile]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]	[Evaluation]